

Spec-Driven Cognitive Scaffolding

Deterministic Architecture for Autonomous Coding Agents

```
[SYSTEM.BOOT] :: SDCS_CORE_v1.1
```

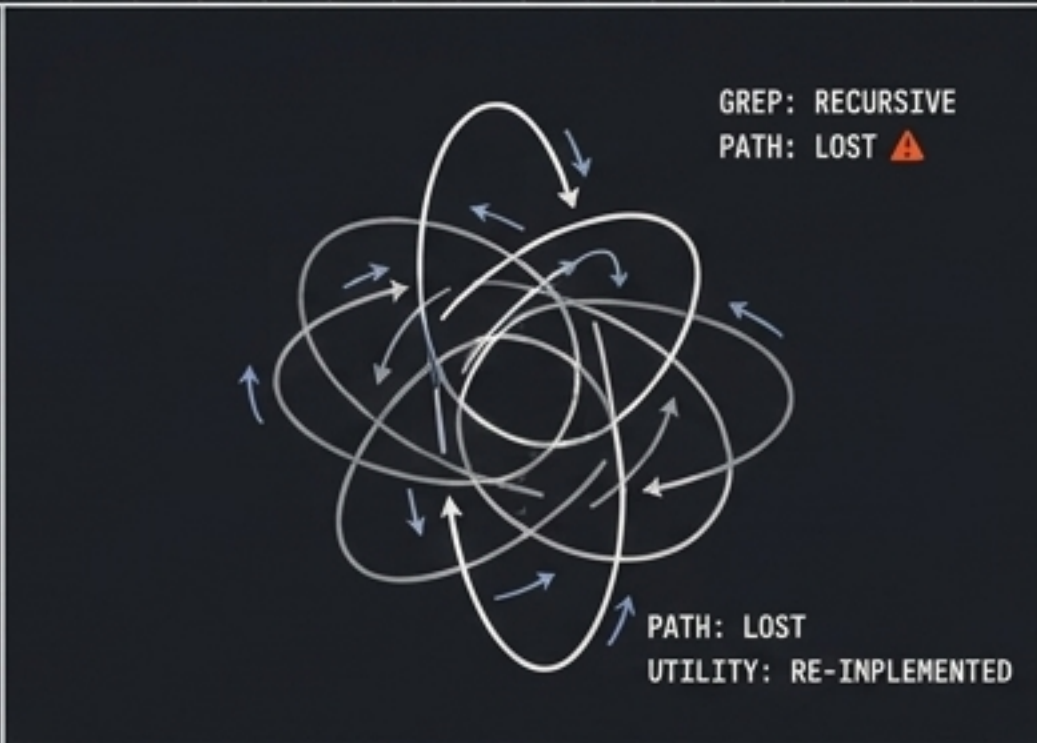
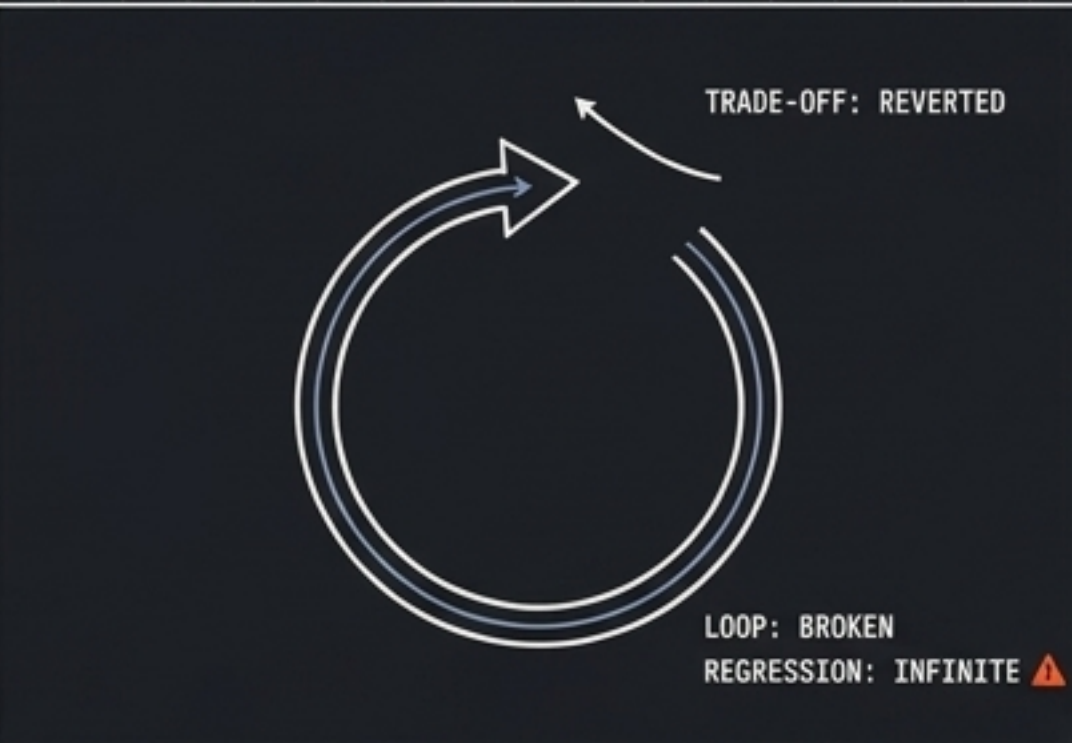
```
[OK] CONSTITUTIONAL INVARIANTS LOCKED
```

```
[OK] DECLARATIVE TOPOLOGY MOUNTED
```

```
[OK] SYMMETRIC EPISODIC MEMORY ONLINE
```

The Failure of Monolithic Prompting

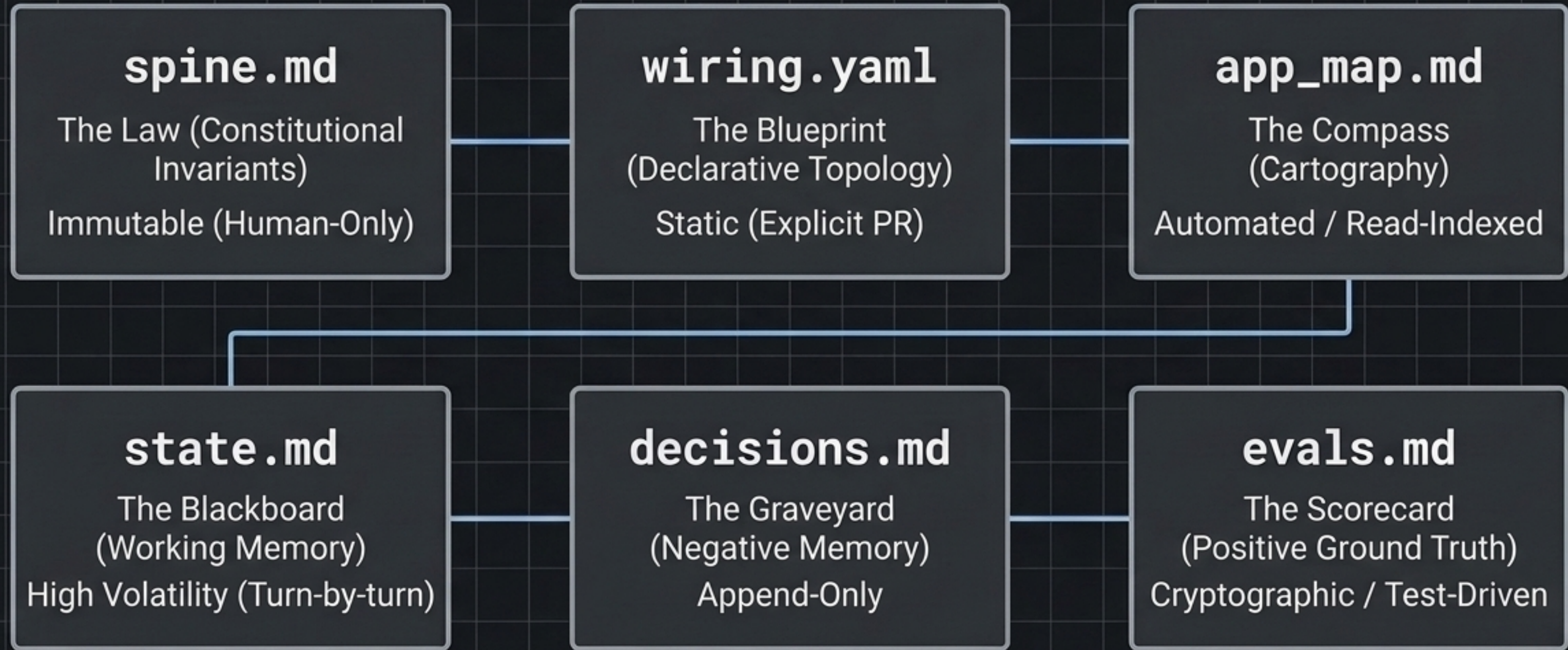
Why Modern Coding Agents Collapse in Real Codebases

 STATE: FADING WEIGHT: <8.01 RESIDUE: ACCUMULATED	 GREP: RECURSIVE PATH: LOST ⚠ PATH: LOST UTILITY: RE-IMPLEMENTED	 TRADE-OFF: REVERTED LOOP: BROKEN REGRESSION: INFINITE ⚠
Context Drift	Cartographic Hallucination	Episodic Amnesia
Core instructions fade from attention weight as multi-turn conversations accumulate debug residue.	Agents execute recursive grep/find loops, lose directory boundaries, and re-implement existing utilities under false paths.	Agents revert intentional trade-offs during bug fixes, creating infinite regression loops.

THE ROOT CAUSE: Treating probabilistic LLMs like **deterministic compilers** instead of **state machines** requiring externalized memory.

The 6-Pillar Architecture

Decoupled Cognitive Subsystems of SPEC-001 v1.1



Pillars 1 & 2: Boundary & Topology

Constraining the operational blast radius

spine.md

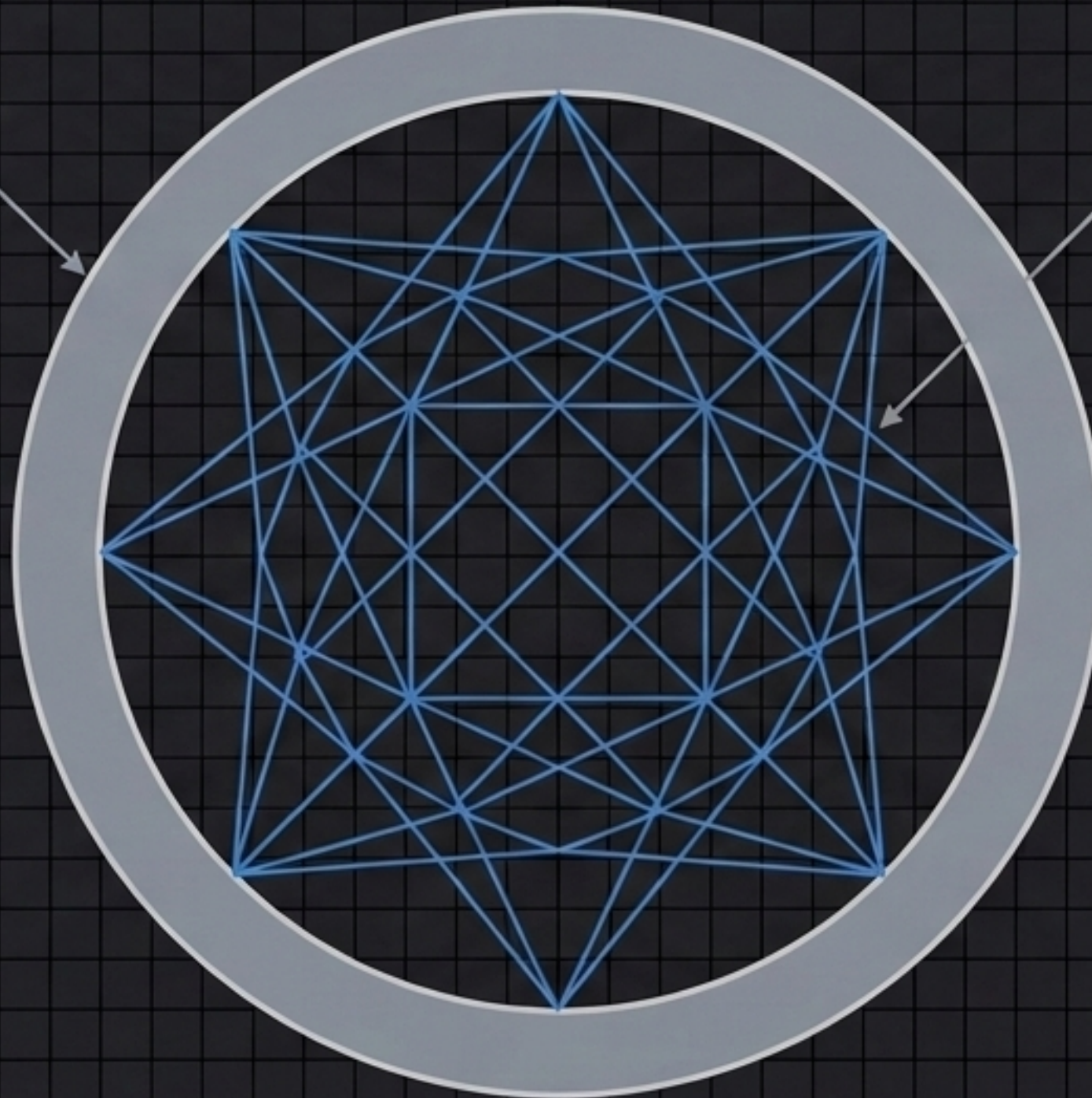
Constitutional Invariants

- Immutable domain axioms that cannot be prompt-negotiated.
- Deterministic verification gates (linters, test suites, coverage).
- Absolute forbidden actions (e.g., unbounded queries, modifying invariants).

wiring.yaml

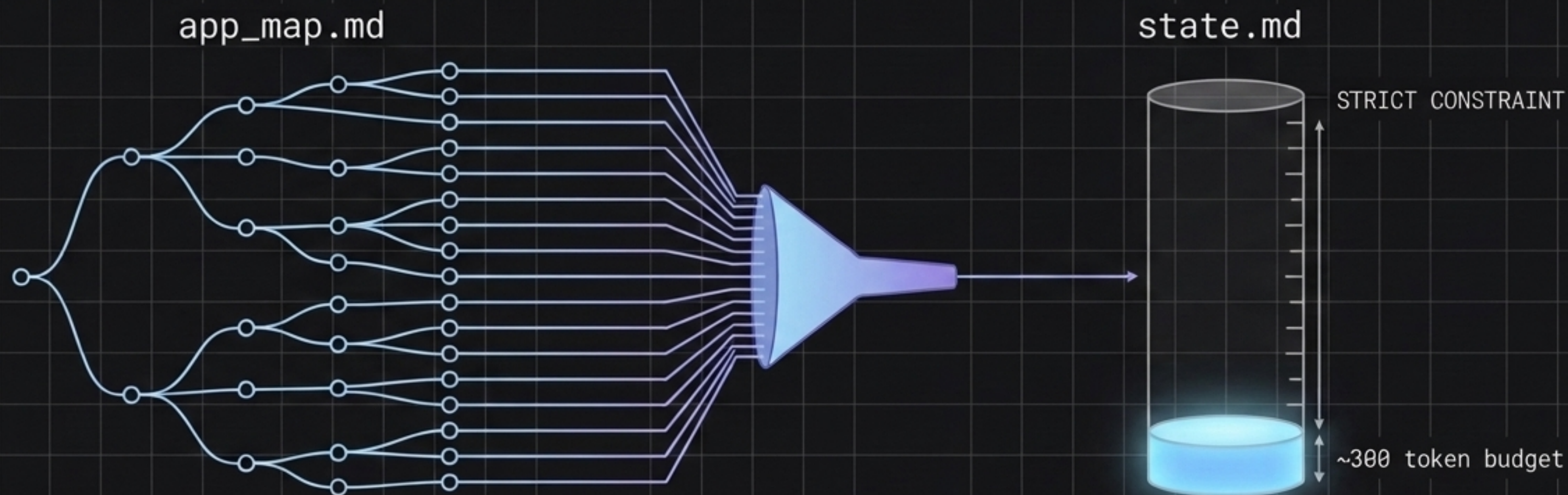
Declarative Topology

- Maps the service mesh and interface bindings.
- Defines tool boundaries and strict environment contracts.
- Mandates explicit inputs and outputs to prevent circular import loops.



Pillars 3 & 4: Cartography & Working Memory

Navigating without token exhaustion



Codebase Cartography

- Complete structural map of modules and export responsibilities.
- The Golden Rule: The agent consults `app_map.md` first, reading only the specific files needed for the task.
- Zero brute-force search loops.

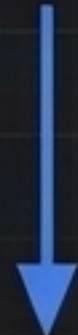
The Blackboard

- Houses ephemeral, turn-by-turn ground truth (active objective, blockers, pending verification gates).
- Strictly pruned after every completed milestone.
- Maintains a ~300 token budget.

Pillar 5: Negative Episodic Memory

Preventing cyclical regression loops across agent sessions

Attempted Fork



Validation Failure (Rejection)



`decisions.md`
(The Graveyard)

MANDATORY 3-PART REJECTION SCHEMA

REJ-014: Morphological Closing Filter on Subfloor Mask

THE CLAIM:

Applying a 5x5 closing filter bridges broken line segments.

THE MEASUREMENT:

Halved room precision from 86.4% → 43.1%; spawned 14 phantom polygons across bathroom fixtures.

WHAT WOULD REOPEN IT:

Blueprints scanned at < 150 DPI where contour continuity drops below 30%.

Pillar 6: Positive Ground Truth & Standing

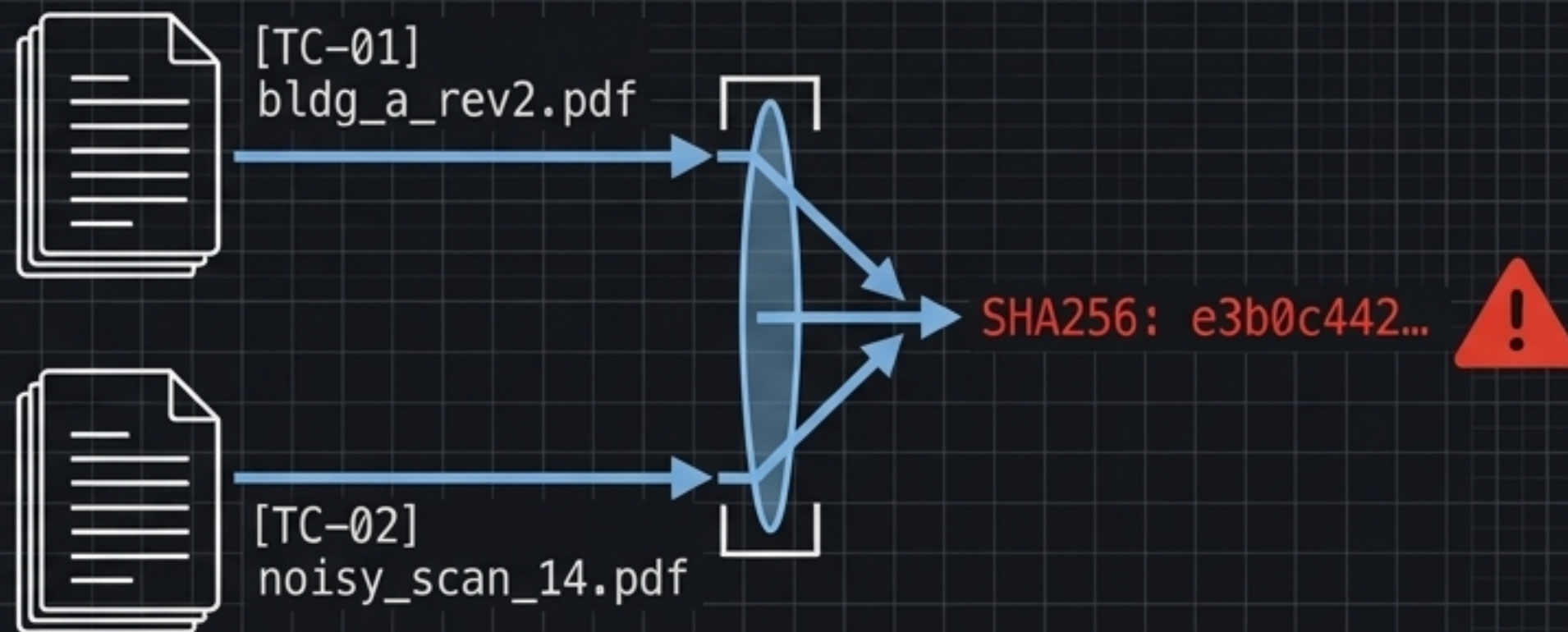
Establishing empirical standing to defeat the Phantom Corpus trap

evals.md

Empirical Standing: Defines exactly what the engine is scored against right now.

The Phantom Corpus Trap: A common agent failure where duplicate templates masquerade as broad test coverage under different filenames.

Corpus Diversity Invariant: Every test fixture is cryptographically fingerprinted via SHA-256 to guarantee unique value.



The Flight Recorder Protocol

Strict separation of working memory and historical audit logs

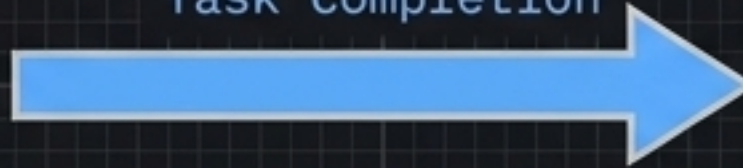
state.md



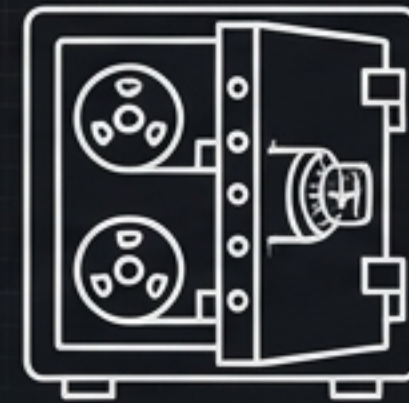
(The Whiteboard)

- Status: Volatile / Ephemeral.
- Actively read on Turn 1 boot.
- Overwritten turn-by-turn.

Task Completion



sessions/*.md



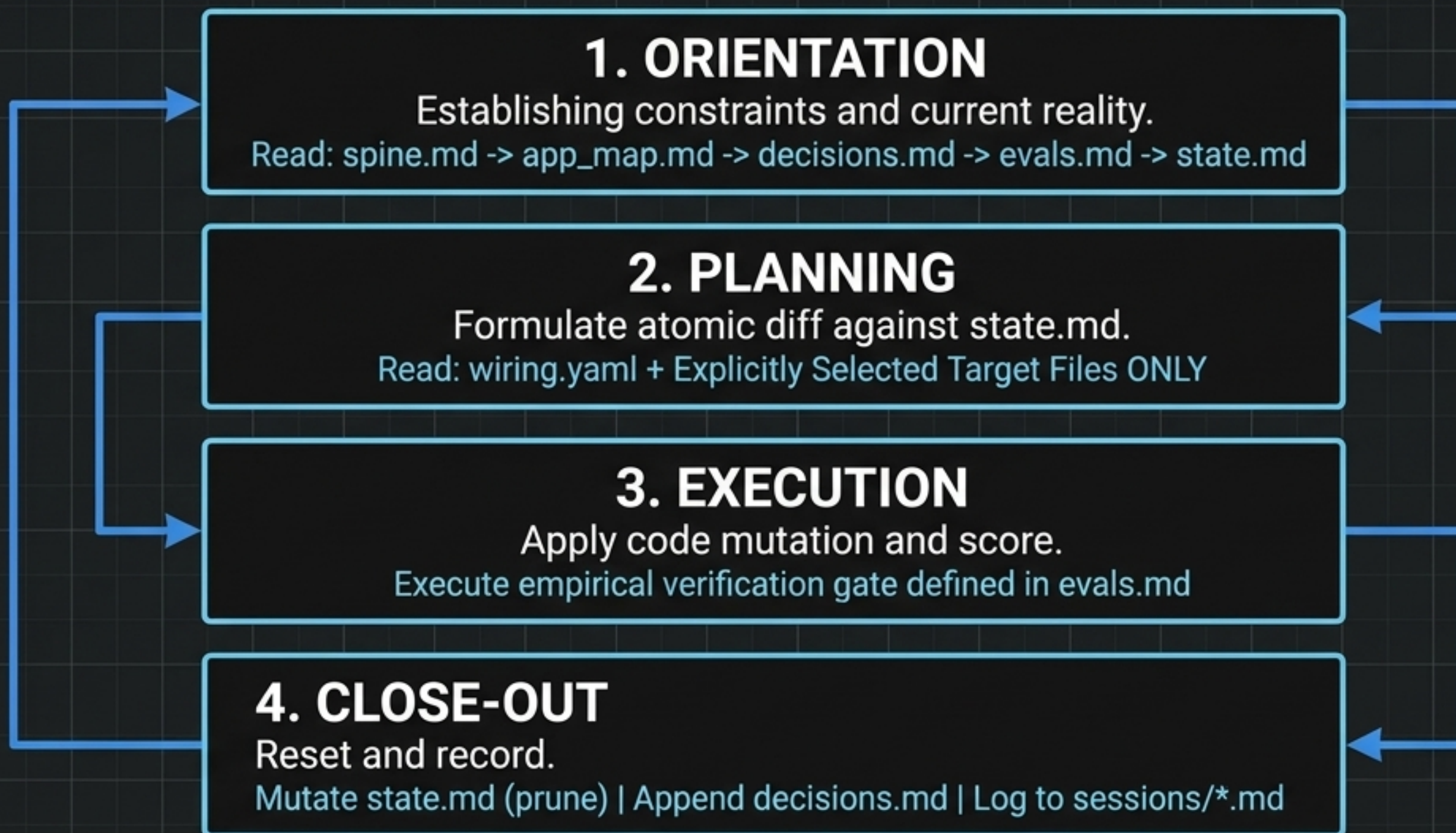
(The Flight Recorder)

- Status: Immutable / Historical.
- Write-once discrete audit log.
- **CRITICAL INVARIANT: NEVER ingested on system boot.**

By physically separating working state from historical logs, SPEC-001 eliminates git merge conflicts and strictly protects context windows from accumulating conversational bloat.

The Autonomous Execution Cycle

SYNTHESIS: Transforming static files into a 4-phase continuous execution engine



Automated Integrity Enforcement

Tooling and deployment scripts to bootstrap the SPEC-001 framework

`sdc_init.py`

- Scaffolds the full 6-pillar framework into any existing repository.
- Automatically indexes existing repository files into `app_map.md`.
- Generates compliant `AGENTS.md` and session templates.



`audit_evals_corpus.py`

- Validates all fixture paths listed in `evals.md` exist on disk.
 - Recalculates SHA-256 hashes to catch unversioned drift.
- ⚠ Automatically halts system execution if duplicate byte-identical files are detected.**